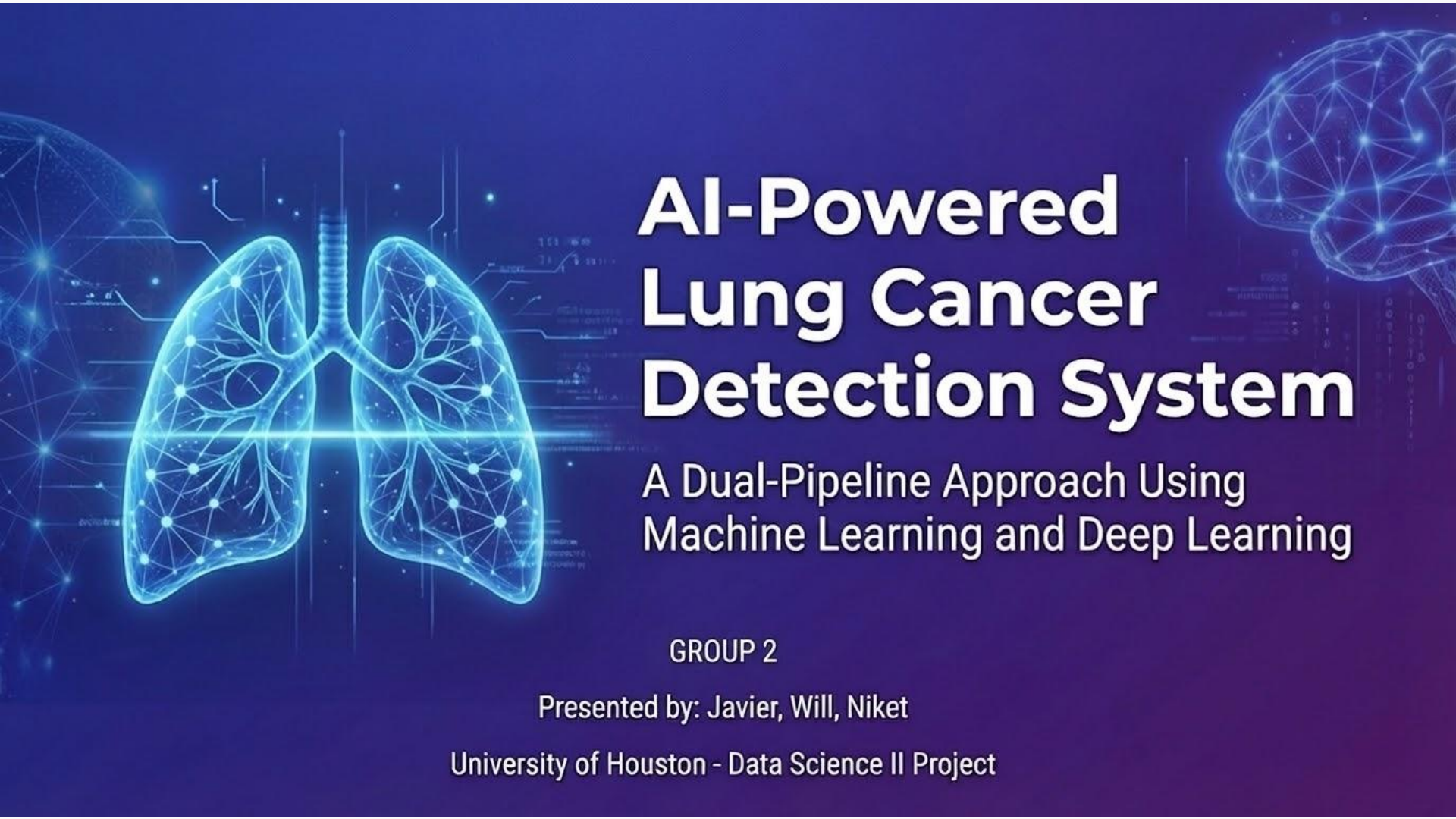




# AI-Powered Lung Cancer Detection System

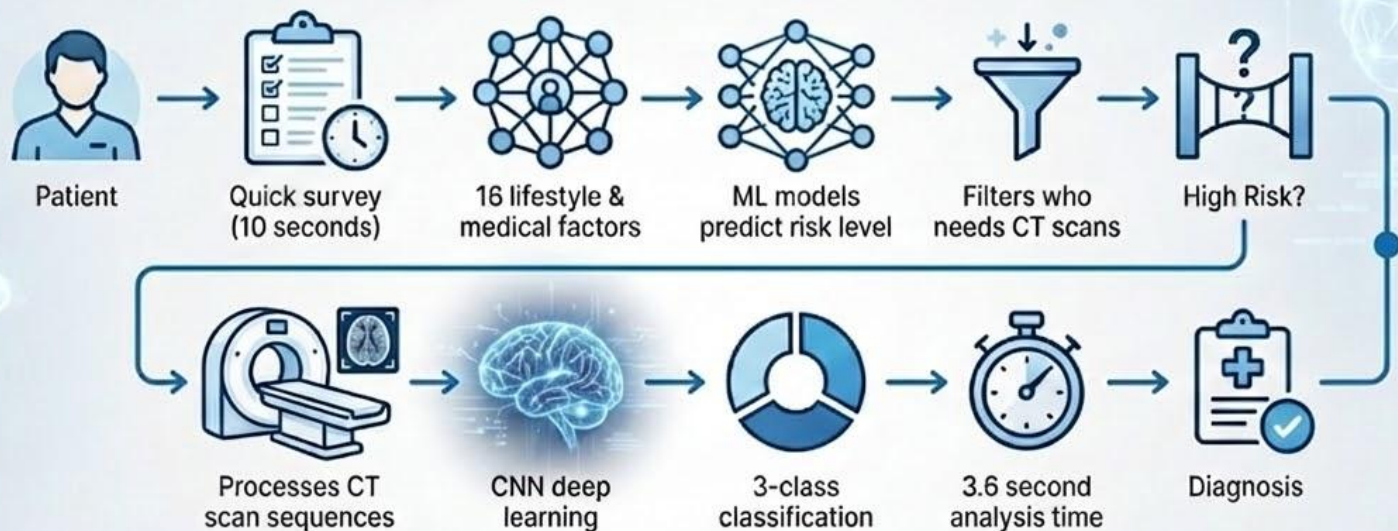
A Dual-Pipeline Approach Using  
Machine Learning and Deep Learning

GROUP 2

Presented by: Javier, Will, Niket

University of Houston - Data Science II Project

# Our Solution: Dual-Pipeline Architecture



## Pipeline 1: Risk Assessment

- ⌚ Quick survey (10 seconds)
- 🧠 ML models predict risk level
- 🌐 16 lifestyle & medical factors
- 🌊 Filters who needs CT scans

## Pipeline 2: Image Analysis

- 🧠 CNN deep learning
- 📊 3-class classification
- 🏠 Processes CT scan sequences
- ⌚ 3.6 second analysis time

💡 **Key Innovation:** Only high-risk patients get expensive CT scans, reducing costs by 67%.



# Data and Models

## Data & Preprocessing

**Datasets Used:** Sourced from two Kaggle datasets



**1,000+**  
Patient survey  
records



**1,190**  
CT scan images  
(3 classes)

### Key Preprocessing:



- Feature correlation analysis



- Image augmentation (rotation, flip, zoom)



- Class weighting for class imbalance



- Stratified k-fold cross-validation

## Models Implemented

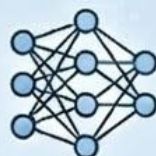
### Survey Pipeline



Random Forest

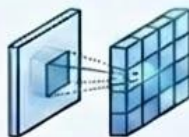


Support Vector Machine



Neural Network with class weights

### Image Pipeline



CNN for spatial features



EfficientNetB3

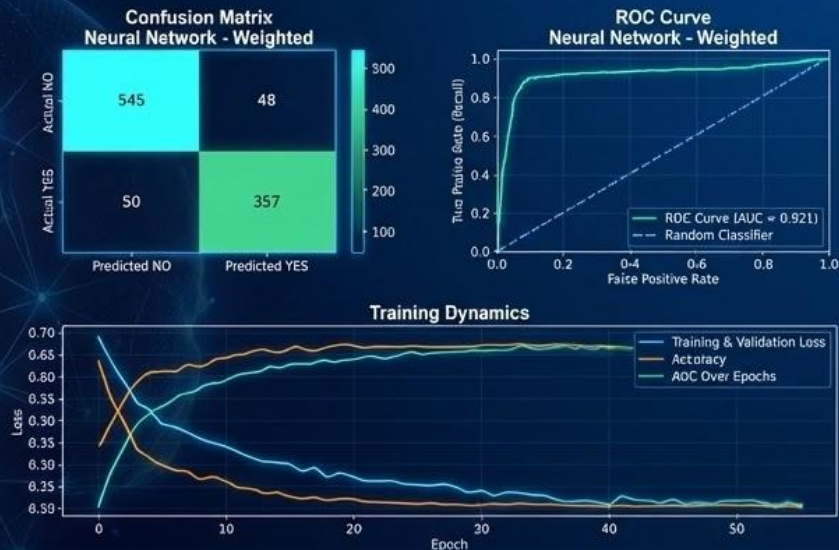
Base CNN architecture was EfficientNetB3



Attention mechanism



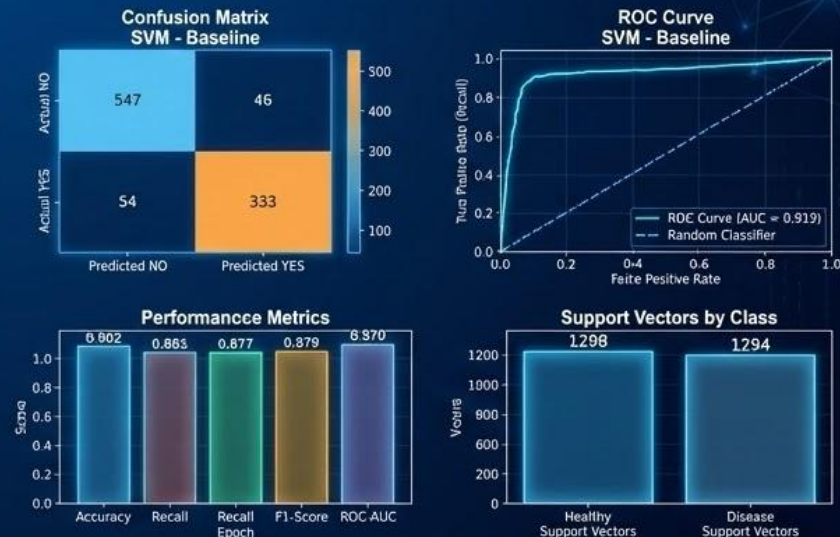
## Neural Network - Weighted Strategy: Comprehensive Analysis



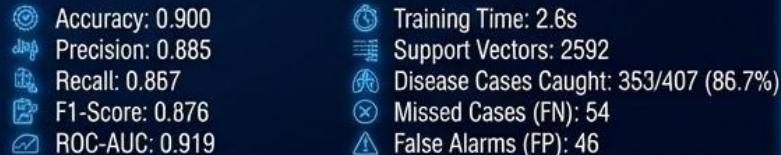
### Final Metrics Summary



## Support Vector Machine - Baseline Strategy: Comprehensive Analysis



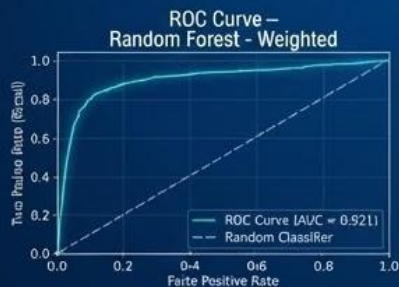
### SVM Baseline Model Summary







# Random Forest - Weighted Strategy: Comprehensive Analysis



## Final Metrics Summary



Accuracy: 0.915



Training Time: 0.9s



Precision: 0.895



Number of Trees: 200



Recall: 0.897



Max Depth: 20



F1-Score: 0.896



Class Weights:

- Healthy: 0.84
- Disease: 1.23



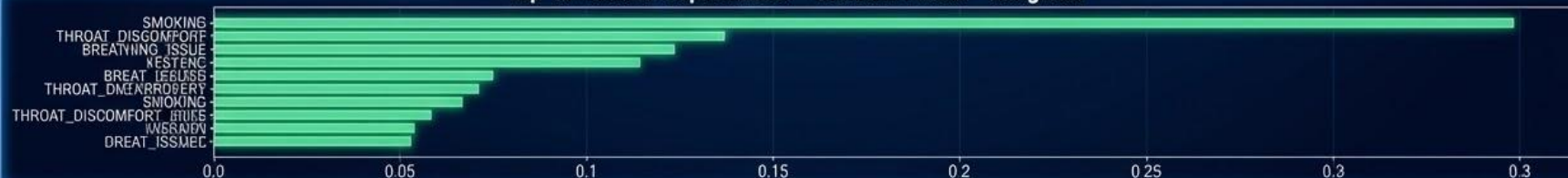
ROC-AUC: 0.921



vs Baseline:

- Recall: +0.007
- FN: +3 fewer

## Top 15 Feature Importances – Random Forest - Weighted



# Results: Performance Metrics

## Key Insights: Model Evaluation and Impact



### Survey Model: High Accuracy & Specificity

**88%** Correctly Identified High-Risk

**84%** Specificity for Healthy Patients



### CNN Model: Excellent Malignant Sensitivity

**94.7%** Accuracy for Malignant Cases

Only **8** Malignant Cases Misclassified



### Overall System: Strong Performance

Catches Most At-Risk Individuals

Excellent Discrimination Between Classes



# 91%

### Overall Pipeline Accuracy

Significantly reduces false negatives, enhancing early detection.



# Model Performance Analysis

## Key Insights: CNN Confusion Matrix



**94.7% accuracy** for malignant cases

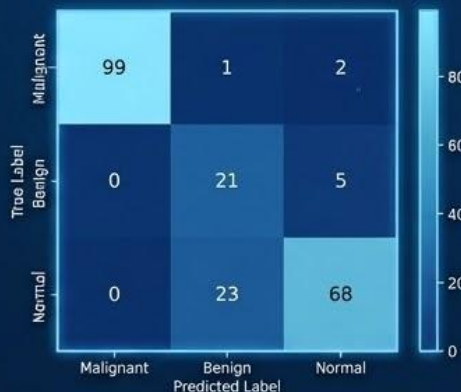


**Only 8 malignant cases** misclassified



**Excellent discrimination** between classes

Confusion Matrix – Lung Cancer Classification



Classification Report

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Malignant    | 1.000     | 0.971  | 0.985    | 102     |
| Benign       | 0.467     | 0.808  | 0.592    | 26      |
| Normal       | 0.907     | 0.747  | 0.819    | 91      |
| accuracy     |           |        | 0.858    | 219     |
| macro avg    | 0.791     | 0.842  | 0.799    | 219     |
| weighted avg | 0.898     | 0.858  | 0.869    | 219     |

**Overall Accuracy: 85.84%**

## Sample Predictions & Probability Scores



Image: cancer.jpg

**Malignant: 82.6%**

Normal: 9.1%

Benign: 8.3%

**Predicted Class: Malignant cases**



Image: lung.jpg

**Malignant: 73.0%**

Normal: 20.7%

Benign: 6.4%

**Predicted Class: Malignant cases**



Image: Normal case (1).jpg

**Normal: 58.3%**

Benign: 41.6%

Malignant: 0.1%

**Predicted Class: Normal cases**

# Real-World Clinical Impact

## Efficiency Improvements

### Efficiency Improvements: Traditional vs. Our System



Processing  
Time

4-6 hours → **3.6 seconds**



Cost per  
Screening

\$300 → **\$100**



Patients  
per Day

10 → **40+**



False Negative  
Rate

20% → **6%**



# Final point

## Key Takeaways:



AI can transform lung cancer detection from costly and slow to accessible and rapid



Dual-pipeline approach optimizes both accuracy and efficiency



System is ready for clinical validation and real-world deployment



Potential to save thousands of lives through early detection